Rehand Abdelhady Yassin

Aerospace Engineer

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Summary

Aerospace Engineer specializing in combustion systems, CFD, and mechanical design. Experience spans propulsion testing, satellite structural analysis at the Egyptian Space Agency, and Formula Student brake systems. Continuing combustion research toward publication.

Education

- **Zewail City of Science and Technology** *Sep 2020 – Jun 2025*
  Bachelor of Science in Aerospace Engineering (Scholarship Recipient) Giza, Egypt

Professional Experience

- **Zewail City of Science and Technology** *2024 – Present*
  Undergraduate Researcher – Combustion Systems
  - Extend the graduation work into research on kerosene–green hydrogen blends, using MATLAB models to evaluate combustion performance and efficiency across equivalence ratios
  - Preparing results for a peer-reviewed publication
- **Zewail City of Science and Technology** *Jul 2025 – Oct 2025*
  Undergraduate Internship Mentor – CNC Foam Cutter
  - Supervised 8 undergraduate students through the design and implementation of a CNC foam cutter system
  - Trained students in stepper motor control, mechanical assembly, Arduino interfacing, and system integration
- **Egyptian Space Agency (EgSA)** *Jul 2025 – Sep 2025*
  Mechanical Design Engineer Intern New Cairo, Cairo, Egypt
  - Contributed to end-to-end satellite design across configuration, structural, thermal, and payload subsystems
  - Designed satellite structural components and mechanical ground support equipment (MGSE) in SolidWorks
  - Performed ANSYS FEA on 8+ structural components to validate integrity under launch loads (shock, vibration, quasi-static)
  - Modeled solar panel deployment kinematics and performed payload sizing in MATLAB
- **Zewail City Racing Team (Velox Team)** *Mar 2023 – Jun 2024*
  Mechanical Team Lead – Brakes and Wheels (Formula Student)
  - Led a sub-team of 5 engineers through design, analysis, and validation of the braking and wheel systems
  - Reduced braking distance by 12% through iterative caliper and rotor sizing
  - Built a MATLAB/Simulink braking dynamics model to predict stopping performance; performed stress analysis in SolidWorks
  - Passed all technical scrutineering checks on the first attempt at the Ever Formula Student competition
- **Zewail City of Science and Technology** *May 2023 – Jan 2024*
  Mechanical Engineering Intern – Pulse Jet Engine Project
  - Contributed to manufacturing, assembly, troubleshooting, and bench testing of a valveless pulse jet engine in a workshop environment
  - Modeled the thermodynamic cycle in MATLAB and validated it against measured thrust data; designed components in SolidWorks
- **EgyptAir Training Academy** *Sep 2023*
  Aircraft Maintenance Intern Cairo, Egypt
  - Rotated through Hangars 6000, 7000, and 8000, studying turbofan, turbojet, and turboprop engine overhaul, gas dynamics, and structural repair
  - Observed fuel, ignition, and landing gear systems; applied maintenance standards and NDT techniques (visual inspection, dye-penetrant testing)
- **Zewail City of Science and Technology** *Sep 2021 – May 2022*
  Junior Teaching Assistant – Mathematics
  - Supported students with problem-solving, assignments, and exam preparation in introductory mathematics
  - Delivered online and in-person tutorials to strengthen core concepts and analytical skills

Projects

- **Kerosene–Hydrogen Combustion Chamber (Graduation Project)** *Jun 2025*
  Zewail City ANSYS Fluent, MATLAB, SolidWorks, Simulink
  - Designed and built a lab-scale test system (blower, preheater, combustor, nozzle) for kerosene, green hydrogen, and blended fuels
  - Ran 15+ ANSYS Fluent cases to evaluate flame stability, temperature distribution, and pressure behavior across fuel compositions
  - Built an interactive web-based digital twin of the test rig
- **NACA 23022 Airfoil CFD Analysis** *Feb 2024 – May 2024*
  Zewail City ANSYS Fluent, MATLAB
  - Performed meshing, setup, and post-processing of lift, drag, pressure, and velocity fields across angles of attack
- **Wind Turbine Blade Design**
  Zewail City MATLAB, QBlade
  - Designed a NACA 23015 blade with chord/twist distribution and off-design analysis; validated with BEM in QBlade
- **Integer Programming for Airline Networks**
  Zewail City MATLAB
  - Built hub location, minimum-cost flow, and routing models applied to Air Cairo's domestic network
- **Mechatronic Linear Servo**
  Zewail City Simulink, Arduino
  - Built hardware, estimated DC motor parameters, and tuned cascaded PID control; validated experimentally
- **Gearbox and V-Belt Shaft Design**
  Zewail City SolidWorks, Inventor
  - Designed a full gearbox assembly and V-belt shaft system with load transfer and structural sizing

Skills

- **CFD & FEA:** ANSYS Fluent (CFD), ANSYS Mechanical (FEA), COMSOL, QBlade (BEM)
- **CAD & Design:** SolidWorks, Autodesk Inventor, Siemens NX, CATIA
- **Programming & Modeling:** MATLAB, Simulink, Python, C++
- **Electronics & Control:** Arduino IDE, Proteus, LTspice, PID control
- **Manufacturing & Testing:** CNC machining, workshop assembly, non-destructive testing (NDT)
- **Languages:** Arabic (Native), English (Fluent), French (Intermediate)

Honors and Awards

- **3rd Place, Youth Climate Mobility Innovation Hackathon (MENA)**, IOM – UN Migration & YOUNGO (Sep 2025): developed Aquavive, an AI-based, energy-efficient water filtration solution for Gaza; presented to a UN/IOM panel
- **2nd Place, Graduation Project Fair**, Zewail City (Jun 2025): lab-scale kerosene–hydrogen combustion chamber
- **Global Nominee, NASA Space Apps Challenge** (Oct 2022): interactive Parker Solar Probe simulation with a QR-based educational game (Unity, GDevelop); top ∼3% of 29,000+ participants
- **4th Place, Medhat Ben Zohair Competition**, Greenish Club (Oct 2022): entrepreneurship concept supporting rural handicraft communities through sustainable tourism

Leadership & Activities

- **Young Negotiator, EGYouth4Climate**, Ministry of Youth and Sports (Present): represent Egypt in climate action initiatives and negotiation simulations; Innovation and Solutions Committee
- **Competitor and Judge, Future Engineers – NASA** (May 2019 – Jun 2023): competed in rover design and space exploration challenges; later served as competition judge
- **Sustainability Ambassador Intern, Be An Ambassador Initiative** (Oct 2024): developed 32 educational videos on sustainability and climate policy
- **Energy Fellow, Student Energy** (Mar–Nov 2024): fellowship in energy systems, project development, and energy transition (Tier 2 track)
- **Participant, McKinsey Forward Program** (Nov 2023 – Apr 2024): training in structured problem-solving, communication, and data-driven decision-making
- **Zewail City events and admissions:** Science Festival organizer / HR / photographer (Jun 2021 – Aug 2023); Admissions Team member (May 2021 – Jan 2022); Commencement organizer (Jul 2023)